Summary Notes:
Azure Storage and Virtual Machines

Overview

Here is an overview of the topics covered in this session.

1. Session 1

a. Azure Storage Types
b. Storage Account Types
c. Replication Strategies
d. Access and Security
e. Blob Storage configuration
f. File Storage Features

2. Session 2

a. VM Planning
b. Virtual Machine Sizing
c. VM storage
d. Connectivity Options
e. Virtual Machine Availability
f. Scaling and Autoscaling
g. Maintenance and Downtime Management

Session 1: Azure Storage Services

1. Azure Storage Types

- **Azure Blob Storage**:
    - Designed for unstructured data such as media files, backups, and logs.
    - Offers three access tiers based on access frequency and cost:
        - Hot Tier: For frequently accessed data, ideal for streaming and real-time data access.
        - Cool Tier: For infrequently accessed data, like backups, with a minimum 30-day retention.
        - Archive Tier: For rarely accessed, long-term archival data, with a minimum 180-day retention.
    - Use Cases: Content delivery (e.g., images, video), big data analytics, archival storage.
- **Azure File Storage**:
    - Provides cloud-based SMB file shares, accessible across platforms, including on-premises.
    - Supports "lift-and-shift" migrations for minimal disruption in file-share environments.
    - Use Cases: Centralized storage for shared company files, application-specific file storage, cross-platform access for teams.
- **Azure Table Storage**:
    - A NoSQL key-value store, schema-less for handling structured, non-relational data.
    - Efficient for applications that need fast access to simple data, like metadata or app configurations.
    - Use Cases: Storage of configuration data, structured logs, metadata, and other key-value data.
- **Azure Queue Storage**:
    - Enables asynchronous messaging between application components, decoupling systems for scalability.
    - Use Cases: Task queue management, message passing in distributed applications, handling backlogged processes without system dependencies.

2. Storage Account Types

- General Purpose v2: Supports blob, file, queue, and table storage with a range of replication and access tiers.

- Blob Storage Accounts: Specialized for blob data only, allowing tiered access to optimize storage costs based on usage patterns.

3. Replication Strategies

- Locally Redundant Storage (LRS): Maintains three copies within a single data center. Low-cost but no geo-redundancy.
- Zone-Redundant Storage (ZRS): Replicates data across three zones within a region, providing redundancy against data center failures within that region.
- Geo-Redundant Storage (GRS): Asynchronously replicates to a secondary region for disaster recovery, suitable for high-availability needs.
- Read-Access Geo-Redundant Storage (RA-GRS): Extends GRS with read access to the secondary region, enabling access even if the primary region fails.
- Geo-Zone-Redundant Storage (GZRS): Combines ZRS with geo-redundancy for ultimate durability, distributing data across multiple zones and regions.

4. Access & Security

- Access Configuration:
    - Every storage object has a unique URL and endpoint, e.g., *`https://<account-name>.blob.core.windows.net/<container>/<blob-name>`*.
    - Custom domains can be set up for easier organization by department or brand.
- Network Security:
    - Firewalls and Virtual Network integration restrict access by IP or subnet.
- Shared Access Signatures (SAS): Grant time-bound, limited access to storage resources without sharing account keys.
- Storage Encryption:
    - Storage Service Encryption (SSE): Default AES-256 encryption secures data at rest.
    - Customer-Managed Keys: For more control over encryption keys, managed via Azure Key Vault.

5. Blob Storage Configuration

- Access Tiers:
    - Automatically tier data to Hot, Cool, or Archive based on access needs to optimize costs.
- Lifecycle Management:
    - Automate data transitions across tiers or delete data to save costs.

- Blob Object Replication:
    - Allows asynchronous replication across different regions to optimize access and enhance availability.

6. File Storage Features

- Snapshots: Incremental backups allow easy restoration from accidental changes or deletions.
- Soft Delete: Retains deleted data for up to 365 days for recovery, enhancing resilience.

Session 2: Azure Virtual Machines (VMs)

1. VM Planning

- Location & Network Configuration:
    - Regions: Select regions based on user proximity to reduce latency and meet compliance needs.
    - Virtual Networks: Create isolated network environments with specific IP ranges and interfaces.
- Naming Standards and Budgeting:
    - Apply standardized naming conventions for easy management and monitor regional pricing for cost efficiency.

2. Virtual Machine Sizing

- Choose VM sizes tailored to workloads, ranging from:
    - General Purpose: Balanced CPU and memory, suitable for diverse workloads.
    - Compute-Optimized: High CPU-to-memory ratio for tasks like batch processing.
    - Memory-Optimized: High memory capacity for databases or large in-memory caches.
    - Storage Optimized and GPU-Optimized: For data-intensive and GPU-based processing respectively.

3. VM Storage

- Disk Types:

    - OS Disk: Stores the operating system, available as SSD or HDD.
    - Temporary Disk: Ephemeral storage for temporary data, reset on VM reboots.
    - Data Disks: Additional storage for application data, offered as Standard or Premium SSD for higher performance.
- Managed Disks: Provides built-in redundancy and simplifies storage management.

4. Connectivity Options

- Bastion: Secure, browser-based RDP/SSH access without exposing public IPs.
- RDP & SSH:
    - RDP for GUI access to Windows VMs (via TCP port 3389).
    - SSH for secure, command-line access to Linux VMs.

5. Virtual Machine Availability

- Availability Sets: Spread VMs across fault and update domains for 99.95% uptime SLA.
- Availability Zones: Offer 99.99% SLA with VMs distributed across separate datacenters within a region, safeguarding against entire data center failures.
- Scaling Approaches:
    - Vertical Scaling: Add more resources (e.g., CPU, RAM) to a single VM.
    - Horizontal Scaling: Add multiple VMs to distribute load, particularly using scale sets.

6. Scaling and Autoscaling

- Virtual Machine Scale Sets: Automatically scales the number of VMs to meet demand, ensuring performance stability.
- Autoscale Configuration:
    - Define minimum, maximum, and default VM counts.
    - Set scaling rules based on CPU or memory usage to control when VMs are added or removed.

7. Maintenance and Downtime Management

- Maintenance Events:
    - Planned Maintenance: Rolling updates to minimize downtime.
    - Unplanned Hardware Maintenance: Uses live migration to handle hardware issues with minimal disruption.

- Unexpected Downtime: Features like automatic healing help ensure continuity by restarting affected VMs.